Mechanics of Human Movement

Course Progress 90.2% (11 of 13)

- Skeletal Muscles: Structure Part- I
- Skeletal Muscles: Structure Part- II
- Mechanics and Modeling of Muscles
- Week 2 - Lecture Notes
- Mechanics of Human Movement : Week 2 Feedback Form
- Quiz: Week 2: Assignment 2
- Week 3
- Week 4
- Week 5
- Week 6
- Week 7

Week 2: Assignment 2

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-08-05, 22:49 IST.

Due date on 2026-08-05, 23:59 IST

1. The ability of a muscle to respond to a stimulation is called 1 Point

○ Extensibility
● Excitability
○ Contractability
○ Responsibility

Yes, the answer is correct. Score: 1
Accepted answer is Excitability

2. Pennate muscles are stronger than parallel muscles. Why? 1 Point

○ Oblique orientation of fibres
○ Presence of more myofibrils
○ Fibre length is less than muscle length
● All of the above

No, the answer is incorrect. Score: 0
Accepted answer is Presence of more myofibrils

3. The finger flexors can make a tight fist when the wrist is in neutral position but not when the wrist is flexed. This is because of 1 Point

○ Geometric constraints
○ Passive insufficiency
● Active insufficiency
○ Eccentric action

Yes, the answer is correct. Score: 1
Accepted answer is Active insufficiency

4. What kind of a muscle is biceps brachii? 1 Point

○ Unipennate
● Parallel fusiform
○ Bipennate
○ None of the above

Yes, the answer is correct. Score: 1
Accepted answer is Parallel fusiform

5. Physiological cross sectional area is defined as 1 Point

● The sum total of all the cross-sectional areas of the fibres in planes perpendicular to the fibre orientation
○ The sum total of all the cross-sectional areas of the fibres in a plane perpendicular to the tendon orientation
○ The sum total of all the cross-sectional areas of the fibres in a plane oblique to the fibre orientation
○ Independent of the cross-sectional areas of the fibres

Yes, the answer is correct. Score: 1
Accepted answer is The sum total of all the cross-sectional areas of the fibres in planes perpendicular to the fibre orientation

6. A combination of actin and myosin makes 1 Point

○ Myofibril
○ Fascicles
● Sarcomere
○ Filament

Yes, the answer is correct. Score: 1
Accepted answer is Sarcomere

7. Tendons possess the following: 1 Point

☑ Viscoelasticity
☑ Non-linear material properties
☐ Flexibility
☐ Plasticity

Partially Correct. Score: 0.66
Accepted answer is Viscoelasticity, Non-linear material properties, Flexibility

8. Which of the following muscle(s) is/are not a part of quadriceps muscle group? 1 Point

☐ Rectus Femoris
☑ Biceps Femoris
☑ Soleus
☐ Vastus Medialis

Yes, the answer is correct. Score: 1
Accepted answer is Biceps Femoris, Soleus

9. The ability of a muscle to lengthen beyond the resting length is known as ____________ 1 Point

○ Elasticity
● Extensibility
○ Spasticity
○ Contracture

Yes, the answer is correct. Score: 1
Accepted answer is Extensibility

10. The biarticular muscle group which can perform both hip extension and knee flexion is ____ 1 Point

● Hamstrings
○ Quadriceps
○ Pectineus
○ Adductor Longus

Yes, the answer is correct. Score: 1
Accepted answer is Hamstrings

11. The amount of force developed is directly proportional to the number of cross bridges formed 1 Point

● True
○ False

Yes, the answer is correct. Score: 1
Accepted answer is True

12. Force production takes place at ________ part of the muscle structure 1 Point

● Myofibril
○ Sarcolemma
○ T Tubule
○ Epimysium

Yes, the answer is correct. Score: 1
Accepted answer is Myofibril

13. In Hill type muscle model, tendons are modelled as 1 Point

○ Contractile component
● Series elastic element
○ Parallel elastic element
○ None of the above

Yes, the answer is correct. Score: 1
Accepted answer is Series elastic element

14. Short contraction of muscle is known as ___________ 1 Point

○ Contracture
● Twitch
○ Contraction
○ Tetany

Yes, the answer is correct. Score: 1
Accepted answer is Twitch

15. The muscle attachment point that typically moves with the contraction and is farther from the midline is known as ________ 1 Point

● Insertion
○ Origin
○ Action
○ Reaction

Yes, the answer is correct. Score: 1
Accepted answer is Insertion